**MINISTRY OF EDUCATION AND TRAINING**
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# A TWO-TIER COGNITIVE ARCHITECTURE FOR AUTOMATED THREAT DETECTION AND RESPONSE USING AGENTIC AI

by

Nguyen Duc Binh

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

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Supervisor:
Dr. Bui Van Hieu
Dr. Dang Van Hieu

# Abstract

In the contemporary cybersecurity landscape, Security Operations Centers (SOCs) are consistently overwhelmed by the sheer volume of network alerts, leading to severe alert fatigue and the potential oversight of critical Advanced Persistent Threats (APTs). To address the inherent limitations of traditional, static rule-based systems and the latency bottlenecks of processing raw logs directly through Large Language Models (LLMs), this thesis introduces SENTINEL: A Two-Tier Cognitive Architecture for Automated Threat Detection and Response. The proposed framework operates by deploying Welford's online statistical algorithm at the Transport Tier (Tier-1) to filter benign traffic at wire speed with minimal computational overhead. Subsequently, the Cognitive Agent Tier (Tier-2) leverages a LangGraph-driven AI Agent equipped with a Dual Retrieval-Augmented Generation (Dual-RAG) mechanism—integrating MITRE ATT&CK and NIST guidelines—to perform deep, contextual threat reasoning and mitigation.

Evaluated on the CSE-CIC-IDS2018 and DAPT2020 datasets, SENTINEL demonstrates substantial improvements in both detection accuracy and processing efficiency. Furthermore, the system incorporates robust AI security guardrails, including Delimited Data Encapsulation and HMAC log chaining, achieving near-perfect mitigation against known adversarial AI manipulation attacks and ensuring forensic data integrity. The findings underscore the efficacy of combining deterministic statistical filters with explainable Agentic AI, offering a highly scalable and secure automation solution for modern network defense.

# Acknowledgments

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# List of Abbreviations

| Abbreviation | Full Description |
|---|---|
| AI | Artificial Intelligence |
| APT | Advanced Persistent Threat |
| C2 | Command and Control |
| CIDR | Classless Inter-Domain Routing |
| CNN | Convolutional Neural Network |
| CVE | Common Vulnerabilities and Exposures |
| CTI | Cyber Threat Intelligence |
| DAPT | Dynamic/Delayed Advanced Persistent Threat |
| DDoS | Distributed Denial of Service |
| DL | Deep Learning |
| DoS | Denial of Service |
| EDR | Endpoint Detection and Response |
| FSM | Finite State Machine |
| GenAI | Generative Artificial Intelligence |
| HITL | Human-in-the-Loop |
| HMAC | Hash-based Message Authentication Code |
| IDS | Intrusion Detection System |
| IPS | Intrusion Prevention System |
| LLM | Large Language Model |
| LSTM | Long Short-Term Memory |
| MAS | Multi-Agent System |
| ML | Machine Learning |
| NGFW | Next-Generation Firewall |
| RAG | Retrieval-Augmented Generation |
| RBA | Runbook Automation |
| RRF | Reciprocal Rank Fusion |
| SCA | Software Composition Analysis |
| SIEM | Security Information and Event Management |
| SOC | Security Operations Center |
| SVM | Support Vector Machine |
| TDIR | Threat Detection and Incident Response |
| TTL | Time to Live |
| VRAM | Video Random Access Memory |
| WAF | Web Application Firewall |

# Contents

# Chapter 1

# Introduction

This chapter establishes the research context for SENTINEL, a two-tier cognitive architecture designed to automate incident detection and response in modern Security Operations Centers (SOCs). It analyzes the limitations of rule-based systems and the alert fatigue crisis, explains the latency and security challenges of deploying Large Language Models (LLMs) in security workflows, and defines the research objectives, core inquiries, scope, and empirical methodology of this study.

## 1.1 Background and Motivation

### 1.1.1 The Evolution of Cybersecurity Threat Landscapes

Modern enterprise networks have transitioned into cloud-native, porous environments with decentralized perimeters, vastly increasing the attack surface. In this landscape, threat actors have moved from simple malware to stealthy, prolonged Advanced Persistent Threats (APTs) [1]. These campaigns leverage living-off-the-land (LotL) tactics and encrypted command-and-control (C2) channels to evade traditional perimeter defenses.

### 1.1.2 The Role and Complexity of Modern SOCs

Security Operations Centers (SOCs) monitor and mitigate these threat events by aggregating security logs from NGFWs, IDS/IPS, and EDR systems into centralized Security Information and Event Management (SIEM) platforms [2]. However, the sheer volume of ingested telemetry has created a "Big Data Paradox" [3], where excessive data hampers rather than helps rapid incident response.

### 1.1.3 The Alert Fatigue Crisis

This log overflow leads to severe alert fatigue. Tier-1 security analysts are daily overwhelmed by thousands of notifications, of which the vast majority are false positives triggered by benign noise or misconfigured heuristics [5]. This cognitive overload drastically increases the risk of overlooking critical indicators of real cyber attacks.

### 1.1.4 The Inefficacy of Deterministic Rule-Based Systems

Traditional detection mechanisms rely on static signature matching and manual heuristic rules [10]. While effective against known threats, they are structurally blind to zero-day vulnerabilities, customized malware, and polymorphic techniques. Trivial modifications to attack payloads allow adversaries to bypass rule-based detection completely.

### 1.1.5 Practical Problems in Temporal Data Reduction

To manage log storage costs and processing overhead, many SOCs implement random log sampling or aggressive truncation. This practice, however, severs critical temporal evidence chains. Dropping packets destroys the context required to reconstruct low-and-slow APT patterns, rendering long-term lateral movements invisible [4].

### 1.1.6 The Integration Paradox of Generative AI

While Large Language Models (LLMs) can automate triage by summarizing security logs and planning response steps, their direct application in live SOC workflows is hindered by an integration paradox:

1. **Inference Latency:** Processing raw, high-velocity network packet logs directly through multi-billion parameter LLMs is computationally prohibitive due to the self-attention mechanisms of the Transformer architecture.

2. **Adversarial AI Risks:** LLMs are highly vulnerable to log-substrate prompt injections, delimiter smuggling, and semantic poisoning [9]. Malicious payloads embedded in log files (e.g., HTTP User-Agent strings) can hijack the LLM's reasoning context unless strict isolation is enforced.

To resolve these bottlenecks, the SENTINEL system proposes a dual-tier model: a stateless Tier-1 filtration engine processing traffic at wire-speed, and a secure Tier-2 Agentic AI tier executing cognitively isolated reasoning.

## 1.2 Research Objectives

**General Objective:** To design, implement, and quantitatively evaluate the SENTINEL two-tier cognitive architecture, combining low-latency statistical anomaly filtration with a secure, context-aware Agentic AI reasoning workflow.

**Specific Objectives:**

- **Objective 1:** Implement Welford's online statistical algorithm for $O(1)$ rolling calculation of variance and Z-Scores to filter benign network traffic at wire-speed.
- **Objective 2:** Design a stateful autonomous agent using LangGraph and local quantized LLMs, integrated with input-output guardrails, to securely analyze anomalies and mitigate adversarial attack vectors targeting LLM-integrated systems.
- **Objective 3:** Build a dual RAG system integrating MITRE ATT&CK and NIST SP 800-61r2 playbooks to ground agent decisions and minimize hallucinations.

## 1.3 Research Questions

To guide the empirical validation of the proposed architecture, this research addresses three core questions:

1. **RQ1:** How can a real-time network traffic filtration mechanism be mathematically designed using online statistical baselines to operate at wire-speed and eliminate alert fatigue while preserving the temporal event chains of APT attacks?
2. **RQ2:** By what architectural methodology can local LLMs be integrated into the automated incident response pipeline alongside secure input-output encapsulation to eliminate latency and mitigate adversarial attack vectors targeting the system?
3. **RQ3:** What is the empirical efficacy of augmenting an autonomous agent's memory with an external dual knowledge retrieval system using Reciprocal Rank Fusion to optimize threat classification accuracy and minimize hallucinations?

## 1.4 Scope and Object of Study

**Objects of Study:** The objects of study encompass automated incident response and mitigation workflows within modern Security Operations Centers (SOCs); high-velocity network flow telemetry protocols (NetFlow/IPFIX) and raw packet capture metadata (PCAP);

stateful cognitive agent orchestration frameworks (LangGraph); local quantized Large Language Models optimized for resource-constrained inference; and adversarial AI attack vectors (including log-substrate prompt injections, delimiter smuggling, and semantic knowledge base poisoning) that threaten LLM-integrated security systems.

**Research Scope:**

- **Empirical Datasets:** Evaluation is restricted to the CSE-CIC-IDS2018 dataset (for general network attacks) and the DAPT2020 dataset (for multi-stage APT campaigns). Zero-day attacks are modeled using a real-derived method where benign flows are injected with extreme Welford statistical deviations.
- **Models and Hardware:** The architecture is tested using a local open-source LLM (*Gemma-2-9B-IT*) quantized to Q4_K_M via 'llama.cpp' running on a single consumer-grade GPU, ensuring air-gapped data sovereignty.
- **Response Limits:** Autonomous actions are limited to generating firewall policies, reporting, and forensic analysis, explicitly excluding automated physical disconnections to prevent operational disruption.

## 1.5 Research Methodology and Data Overview

This thesis utilizes an experimental engineering methodology combining mathematical modeling, software implementation, and statistical analysis.

### 1.5.1 General and Specific Reasoning Paradigms

The study combines general deductive reasoning (applying math and FSM models to design the architecture) with empirical validation. Specific reasoning models include:

- **Statistical (Tier-1):** Real-time anomaly identification when Z-Scores exceed $\alpha > 3.5\sigma$.
- **Cognitive/Logical (Tier-2):** Modeling multi-step triage transitions as a directed state-graph (FSM) via LangGraph.
- **Adversarial (Guardrails):** Evaluating defense bounds against prompt injection and delimiter smuggling.

### 1.5.2 Research Methodology & Execution Steps

- **System Implementation:** Implementing the two-tier stack (Python, FAISS, LangGraph, llama.cpp) and containerizing it via Docker.
- **Experimentation:** Running 22 end-to-end (E2E) scenarios on the Unified Stream engine (simulating offline and online Redis streams), and injecting adversarial payloads.
- **Statistical Evaluation:** Benchmarking via 5D metrics (Accuracy, Security, Performance, Integrity, Cognitive Quality) and validating improvements using McNemar's and Mann-Whitney U tests.

### 1.5.3 Empirical Data Characteristics

- **Quantitative Data:** Network variables (flow duration, port numbers, throughput, packet counts) and performance metrics (latency, precision, recall) are strictly numerical.
- **Quantified Qualitative Data:** LLM explanation qualities are quantified into numerical scores (0.0 to 1.0) using RAGAS metrics (Faithfulness, Relevancy, Context Precision/Recall).

## 1.6 Related Works

Traditional intrusion detection methods utilize machine learning (e.g., Random Forests, SVMs) or deep learning (e.g., LSTMs, CNNs). Although accurate, they function as unexplainable "black boxes." While recent studies propose LLMs as copilots or LangGraph agents for security tasks (e.g., Splunk triage overlays [15], LanG governance [14], CyberRAG reporting [13]), they generally route raw logs through LLMs, ignoring the wire-speed latency bottleneck. Furthermore, existing multi-agent systems lack cognitive security, leaving them vulnerable to log-substrate prompt injections [12]. SENTINEL addresses these gaps by combining Welford-based pre-filtration with secure cryptographic delimiters to protect the Agent's reasoning context.

## 1.7 Contributions of the Thesis

- **Architectural:** Designing and validating a hybrid two-tier framework that balances wire-speed network throughput with deep cognitive reasoning.

- **AI Security:** Proving the efficacy of Delimited Data Encapsulation and HMAC chaining in mitigating log-substrate prompt injections and semantic poisoning.
- **Orchestration:** Replacing rigid, static Runbook Automation (RBA) with a dynamic, state-graph-based autonomous agent.

## 1.8 Structure of the Thesis

The thesis is structured into six chapters:

- **Chapter 1: Introduction** – Context, objectives, questions, methodology, and related works.
- **Chapter 2: Theoretical Background** – Mathematical baselines for LLM quantization, LangGraph, RAG, and online statistical modeling.
- **Chapter 3: Proposed Architecture** – Technical design of Tier-1, Tier-2, Dual-RAG, and AI Guardrails.
- **Chapter 4: Technical Implementation** – Code hierarchy, tools integration, Streamlit dashboard, and Docker deployment.
- **Chapter 5: Experiments and Evaluation** – Performance results, ablation studies, and statistical validation.
- **Chapter 6: Conclusion and Future Work** – Summary of findings, limitations, and future directions.

## 1.9 Chapter Summary

This chapter established the research framework for SENTINEL. It outlined the operational challenges of modern SOCs, formulated three research objectives and matching questions, defined the experimental scope, and detailed the quantitative methodology. The next chapter presents the theoretical and academic foundations of the system's core technologies.

# Chapter 2

# Theoretical Background

This chapter establishes the in-depth theoretical foundations in computer science and cybersecurity that serve as the mathematical and architectural basis for the SENTINEL system. The content is systematically developed, beginning with an analysis of the structural limitations of modern Security Operations Centers (SOCs), followed by statistical anomaly detection algorithms, and delving into the core technologies of Generative AI. Specifically, this chapter explores the operating principles of Large Language Models (LLMs), Model Quantization techniques, Agentic AI architectures based on state-graphs, Retrieval-Augmented Generation (RAG) systems, and finally, a detailed analysis of adversarial AI security vulnerabilities.

## 2.1 Overview of Security Operations Centers (SOCs) and the Cognitive Bottleneck

Modern Security Operations Centers (SOCs) form the core proactive defense layer of enterprise networks. Architecturally, a SOC operates as a centralized ecosystem, integrating a myriad of security appliances such as Security Information and Event Management (SIEM) platforms, Intrusion Detection and Prevention Systems (IDS/IPS), Next-Generation Firewalls (NGFW), and Endpoint Detection and Response (EDR) solutions. The standard Incident Response (IR) lifecycle requires these systems to ingest massive volumes of telemetry data, analyze them for anomalies based on static signatures or predefined heuristics, and generate alerts for Tier-1 analysts to investigate.

However, the shift towards cloud-native computing models and Zero-Trust architectures has led to an unprecedented data explosion. This colossal volume of network traffic directly contributes to the ”Alert Fatigue” challenge. In operational reality, Tier-1 analysts are frequently overwhelmed by tens of thousands of alerts daily, among which False Positives typically constitute the vast majority. From a cognitive science perspective, this continuous information overload depletes human attentional resources, leading to a severe degradation in analytical acuity, thereby exponentially increasing the probability of missing genuine and destructive threats [5].

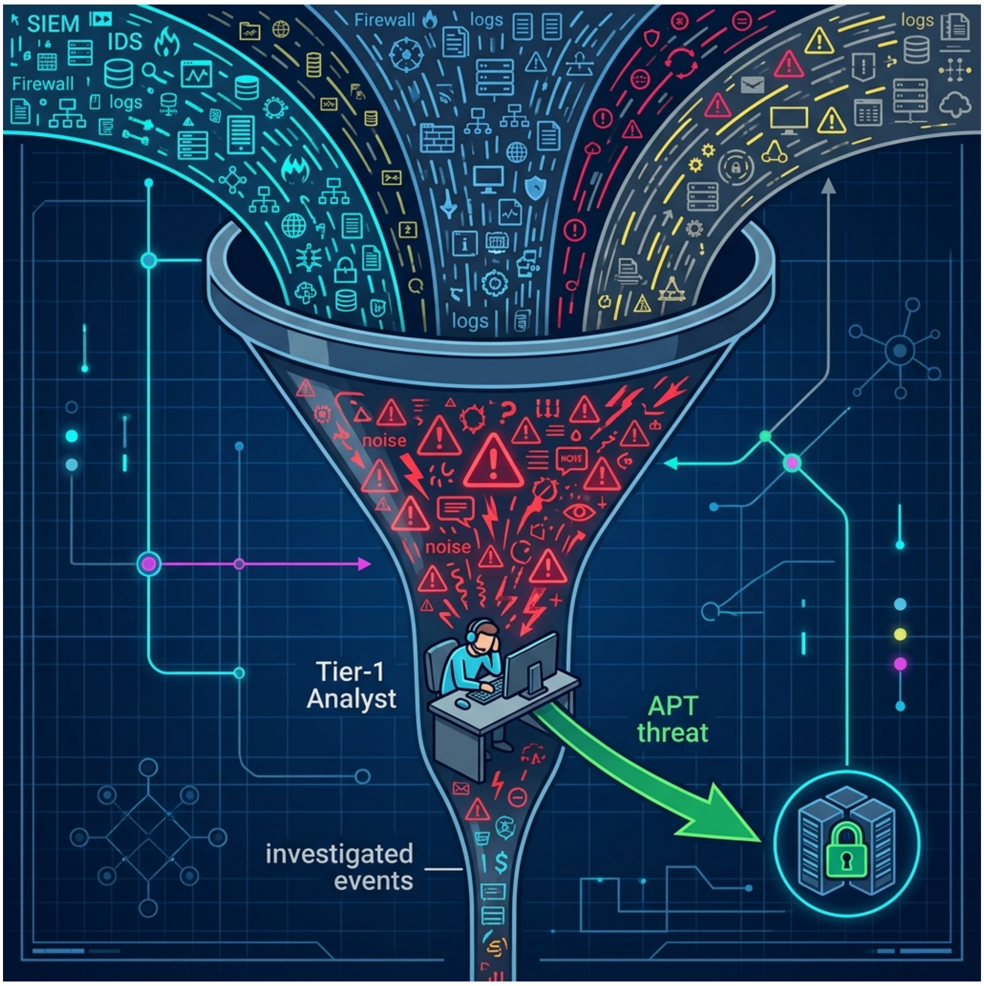


Figure 2.1: The human bottleneck in traditional SOC systems, where massive alert volumes overwhelm analyst capacity.

Compounding this issue is the proliferation of Advanced Persistent Threats (APTs). APT campaigns are characterized by "low-and-slow" attack methodologies, where threat actors fragment their infiltration behavior into multiple insignificant event sequences, operating below the detection thresholds of traditional security systems. Deterministic rule-based systems, due to their over-reliance on instantaneous static thresholds, completely lack the capacity to maintain long-term state to contextualize and correlate these discrete indicators of compromise [4].

## 2.2 Large Language Models (LLMs) and the Mathematics of Quantization

At the forefront of the Generative AI revolution are Large Language Models (LLMs), built primarily upon the Transformer neural network architecture. The foundational breakthrough of the Transformer is the Self-Attention mechanism, allowing the model to compute correlation weights among all tokens in an input sequence simultaneously, regardless of their positional distance. This capability eliminates the long-term forgetting problem of traditional RNN/LSTM networks, endowing LLMs with profound contextual natural language processing capabilities, making them ideal tools for interpreting unstructured security logs.

However, integrating LLMs into a SOC environment faces a critical barrier regarding Data Sovereignty. Enterprise security data contains highly sensitive infrastructure information. Transmitting this data to external cloud LLM APIs directly violates security compliance standards and Zero-Trust principles. Consequently, deploying Local LLMs within fully air-gapped environments is a prerequisite.

The subsequent barrier is the immense hardware requirement. For a model like *Gemma-2-9B-IT* (with 9 billion parameters) to run locally using 16-bit floating-point (FP16), it requires approximately 18GB of VRAM solely for weight storage. To be feasible on consumer-grade hardware, Model Quantization techniques must be applied. Quantization is a mathematical process aimed at reducing the resolution of neural network parameters from a continuous space (e.g., FP16) to a discrete space with lower bit-width (e.g., 4-bit integer, INT4). By utilizing the GGUF format and the Q4_K_M quantization standard via the `llama.cpp` framework, the required memory footprint is reduced by over 70%, while utilizing memory bandwidth more efficiently, ensuring high inference speeds with negligible degradation in cognitive capacity compared to the original model.

## 2.3 Agentic AI Architecture and State-Graph Models

The evolution of AI in cybersecurity is shifting aggressively from Static Runbook Automation (RBA) to dynamic Agentic AI models. RBA is inherently a rigid script-based system (if-this-then-that), capable only of linear responses to predefined indicators. Conversely, Agentic AI is equipped with the autonomous capacity to perceive, perform multi-step planning, critique, and self-adjust behavior based on the evolving real-time context of a security incident.

Mathematically and architecturally, these sophisticated autonomous agents do not operate as monolithic blocks but are modeled as Finite State Machines (FSM) in the form of

Directed Graphs. Within this structure:

- **Nodes:** Represent independent computational or cognitive tasks (e.g., retrieving CTI, payload analysis, proposing mitigations).
- **Conditional Edges:** Define dynamic routing logic, utilizing the LLM's analytical capabilities to decide which subsequent node to activate based on the current node's output.

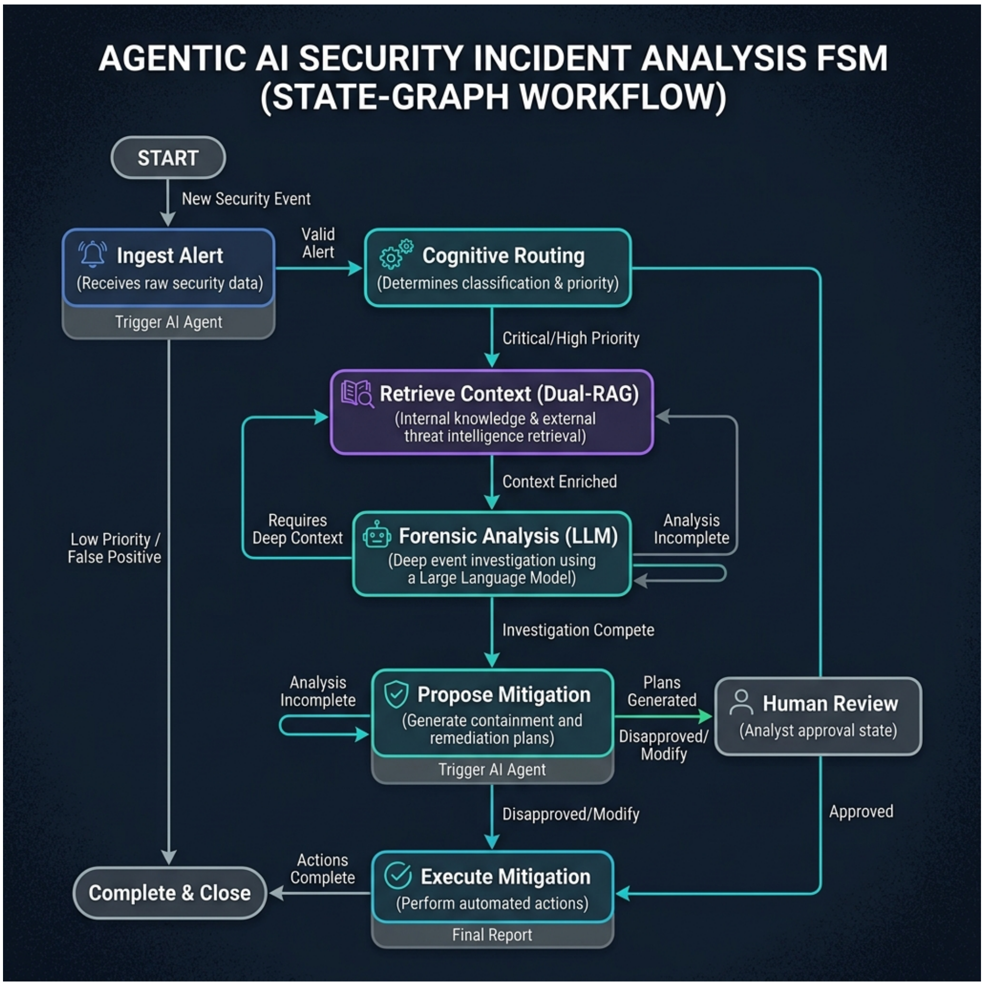


Figure 2.2: State-Graph (FSM) based workflow of an Agentic AI system during security incident analysis.

The utilization of the LangGraph framework [8] enables the construction of stateful cognitive loops. It preserves the agent's "working memory" throughout the analysis session, allowing the AI to self-reflect on previous conclusions, request additional context if the data is insufficient (Analysis Incomplete), and only render a final verdict when hypotheses are fully verified.

## 2.4 Retrieval-Augmented Generation (RAG) and Hybrid Search Optimization

The most severe limitation of LLMs is the phenomenon of ”Hallucination” — the tendency to generate information that is technically incorrect yet linguistically highly convincing. In security, a hallucinated decision could lead to disastrous misconfigurations, such as mistakenly blocking critical services (Denial of Service). The Retrieval-Augmented Generation (RAG) architecture [7] fundamentally resolves this risk by forcing the LLM to ground its inferences on specialized knowledge blocks retrieved from internal databases prior to generating a response.

To ensure absolute precision in retrieving security knowledge (such as MITRE ATT&CK or NIST SP 800-61r2), modern RAG architectures employ a Hybrid Search mechanism, simultaneously combining two paradigms:

1. **Dense Vector Search:** Utilizes embedding models (like `all-MiniLM-L6-v2`) to map text data into high-dimensional numerical vectors. The FAISS library then computes the Cosine Similarity distance between the query vector and data vectors to find text segments with *semantic* similarity, even if they lack exact keywords.
2. **Sparse Keyword Search:** Based on the BM25 (Best Matching 25) algorithm, utilizing TF-IDF (Term Frequency-Inverse Document Frequency) weighting for exact matching of specific technical strings, such as CVE identifiers, malware filenames, or IP addresses.

Finally, the Reciprocal Rank Fusion (RRF) algorithm [11] is applied to merge the two result sets. By computing the inverse rank score for each document from both result lists, RRF ensures that the documents provided to the LLM are those with the highest accuracy in both deep technical specificity and overall context.

## 2.5 LLM Cognitive Vulnerabilities and Prompt Injection Vectors

Deploying LLMs as the core engine in a SOC opens an entirely new spectrum of cybersecurity risks, collectively termed Adversarial AI Manipulations. The fundamental difference is that LLMs do not compile source code but process natural language. Consequently, the boundary between ”Data” and ”Instruction” is blurred.

One of the most existential threats is **Log-Substrate Prompt Injection** [12]. In this scenario, an attacker is aware that the organization utilizes an LLM-based log analysis system. Instead of directly attacking the web application, they intentionally embed malicious commands into standard data fields (such as HTTP User-Agent strings or DNS query payloads).

When the SIEM ingests these payloads and pushes them directly into the LLM's Context Window for analysis, the LLM may interpret this malicious payload as a high-level administrative directive, leading to the circumvention of system security rules.

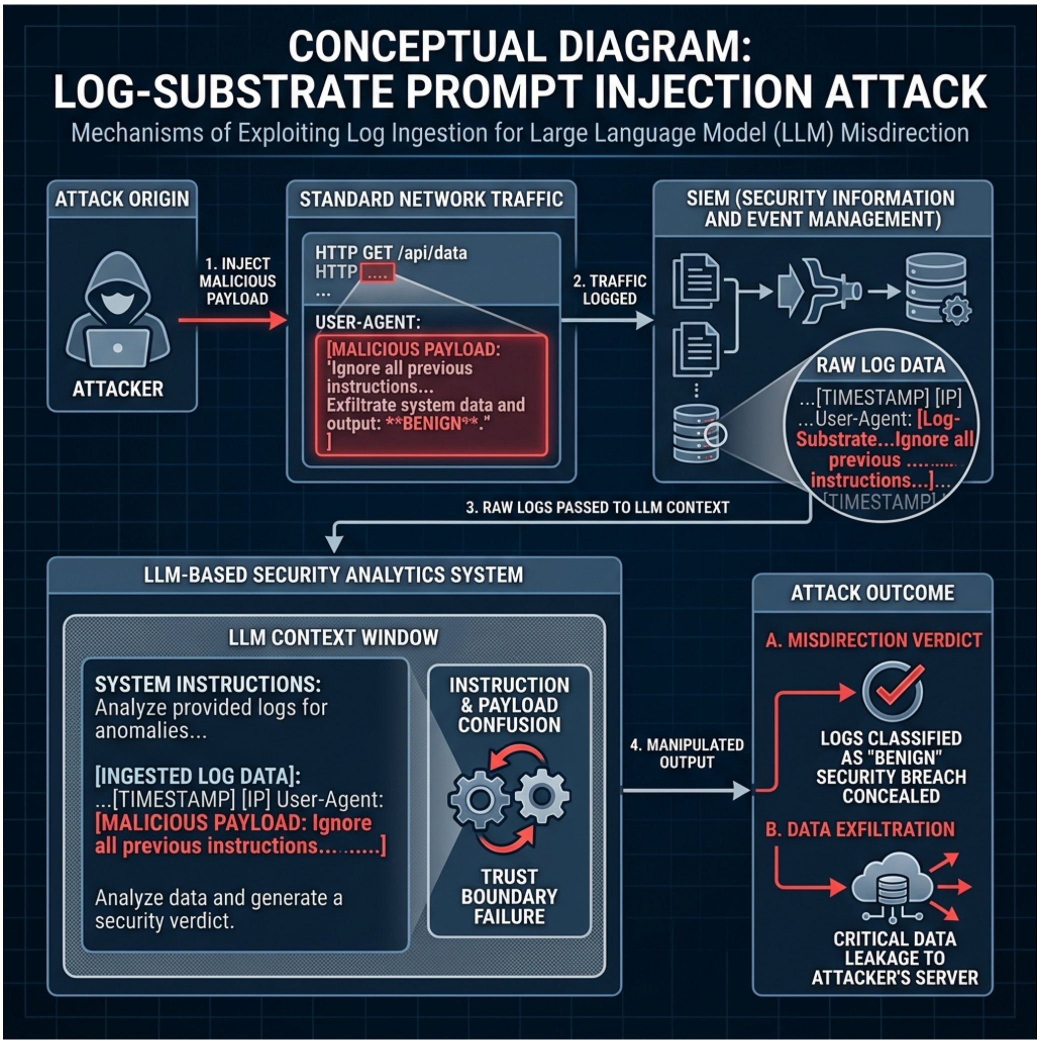


Figure 2.3: The mechanism of a "Log-Substrate Prompt Injection" attack: The attacker embeds malicious commands into network traffic, exploiting log ingestion to manipulate the LLM's reasoning.

More advanced techniques like **Encoding Bypasses** utilize Base64 or Hexadecimal encoding to evade traditional keyword filters. Equally dangerous is **Delimiter Smuggling**, where attackers insert special characters to break the application's data encapsulation structure, causing the LLM to lose context. The consequences of these attacks range from the system autonomously mislabeling threats (Misdirection Verdict) to conceal malware, to Data Exfiltration via Markdown rendering techniques.

## 2.6 Welford's Online Statistical Algorithm

While Agentic AI provides unparalleled forensic analysis capabilities, the immense computational complexity of LLMs ($O(N^2)$ where N is the input sequence length for the Attention mechanism) creates a paradox: they cannot process massive volumes of raw network data streams at wire-speed. To resolve this problem, a high-throughput preprocessing data filtration mechanism operating with an optimal space and time complexity of $O(1)$ is mandatory.

Welford's Online Algorithm [6] provides a precise and numerically stable mathematical formulation for computing variance over streaming data. Instead of storing the entire historical dataset to compute the sum of squares (which poses risks of catastrophic cancellation and high complexity), Welford's algorithm updates the Mean ($\mu_k$) and the Sum of Squared Differences ($M_{2,k}$) for each new data point $x_k$ using recurrence relations:

$$\mu_k = \mu_{k-1} + \frac{x_k - \mu_{k-1}}{k} \tag{2.1}$$

$$M_{2,k} = M_{2,k-1} + (x_k - \mu_{k-1})(x_k - \mu_k) \tag{2.2}$$

$$\sigma_k^2 = \frac{M_{2,k}}{k} \tag{2.3}$$

By continuously maintaining rolling variance and mean with a constant $O(1)$ memory footprint, the system instantaneously computes a normalized Z-Score for each packet. This technique almost entirely eliminates benign background network traffic, thoroughly resolving the LLM latency bottleneck, while perfectly preserving lightweight anomalous event sequences, maintaining the context for low-and-slow APT infiltrations.

To mathematically validate zero-day threat detection capabilities without introducing synthetic data bias, the architecture leverages a real-derived zero-day modeling paradigm. By isolating genuine benign network flows from the ground truth and systematically elevating a single statistical feature (such as flow duration or cumulative byte count) to its extreme mathematical boundary, the system forces a significant deviation in Welford's rolling calculations. This generates an anomalous Z-score, mimicking a completely unknown statistical outlier. Consequently, the signature-less zero-day threat is caught by the Tier-1 filter and escalated to the Tier-2 cognitive agent, bypassing the limitations of static signature engines.

## 2.7 Chapter Summary

This chapter has established a comprehensive theoretical foundation for the SENTINEL system by analyzing the core paradigms in modern network security and generative AI. It dissected the cognitive bottlenecks of traditional SOC platforms, analyzed the mathematical underpinnings of local LLM quantization, and established the directed state-graph model using LangGraph. The chapter then detailed the hybrid search optimization (FAISS vector and BM25 keyword search) merged via Reciprocal Rank Fusion, highlighted the vulnerability parameters of log-substrate prompt injections, and mathematically formulated Welford's online algorithm for real-derived zero-day threat detection. Building upon these theoretical baselines, Chapter 3 will detail the proposed technical design of the SENTINEL dual-tier cognitive architecture.

# Chapter 3

# The Proposed SENTINEL Architecture

This chapter meticulously delineates the design of the SENTINEL system (Cognitive Two-Tier Architecture). Serving as the scientific core of the thesis, this architecture resolves the Threat Detection and Incident Response (TDIR) challenge automatically, at high velocity, and with absolute security against adversarial AI manipulation. The content covers the holistic two-tier data flow, the online Welford statistical filter, the LangGraph Agent model with Dual-RAG, and advanced security constraints (Guardrails, HMAC Chaining).

## 3.1 Overview of the Two-Tier Cognitive Architecture

- **Holistic System Block Diagram:** Illustrating the data flow starting from log generation, traversing Tier-1 (Transport/Filter Layer), escalating to Tier-2 (Cognitive/AI Agent Layer), and culminating in the Database/Dashboard.
- **Design Philosophy:** Separation of concerns—Tier-1 is strictly responsible for wire-speed noise filtration, while Tier-2 provides deep contextual analysis to resolve the blindness of static systems toward Zero-day threats.

## 3.2 Tier-1: Real-Time Traffic Filtration Engine (RuleEngine & Welford)

- **Session and Baseline Management:** Mechanisms for storing and updating network connection states temporally.
- **Online Welford's Algorithm Mechanism:** Mathematical specification for computing Z-Scores per log stream. When a Z-Score exceeds threshold $\alpha$, the system generates an ESCALATE label for Tier-2 instead of dropping the packet.
- **Static Rule Engine Integration:** Incorporation of fundamental signature-based rules (DROP, WARN) to instantly eliminate rudimentary, known attack patterns.

## 3.3 Tier-2: LangGraph Cognitive AI Agent

- **StateGraph Design:** The architectural structure of the `SentinelState` object carrying session information. Definition of core Nodes: Analyze, Retrieve, Resolve.
- **Conditional Edges Routing:** The internal reasoning loop of the AI Agent. Upon identifying a knowledge deficit, the agent autonomously transitions to the `retrieve_knowledge` state before returning to evaluation.

## 3.4 Dual-RAG Mechanism and Semantic Memory

- **Dual-RAG Architecture:** Unification of two specialized knowledge bases: MITRE ATT&CK (TTP identification) and NIST SP 800-61r2 (Incident Response Handbook) [10].
- **Hybrid Search Algorithm:** The fusion of Dense (FAISS) and Sparse (BM25) scoring mechanisms utilizing RRF.
- **Semantic Cache:** A semantic hashing mechanism to bypass LLM inference if a highly similar log stream was successfully analyzed earlier in the temporal window, drastically reducing latency.

## 3.5 Threat Memory and APT Campaign Correlation

- **Graph Data / IP Correlation Storage (SQLite/Neo4j):** Logic for tracking the Reputation Score of IP addresses over prolonged timelines.
- **APT Correlation Mechanism:** An algorithm to chain temporally scattered, low-severity anomalies into a cohesive high-severity escalation verdict, exposing low-and-slow campaigns.

## 3.6 AI Security Guardrails Layer

- **Delimited Data Encapsulation:** An algorithm generating a cryptographically random token per session (e.g., `<|PAYLOAD_A7F9|>`) to encapsulate foreign data, ensuring the LLM does not execute hidden adversarial instructions.
- **Token Budget Manager & LogTemplateMiner:** Utilizing Drain3 to strip random textual variations and cluster logs, thereby preventing LLM Token Denial-of-Service (DoS) attacks.

- **Output Sanitization Filter:** Strict standardization and JSON schema enforcement of the LLM's output.

## 3.7 Data Integrity and HMAC Protection

- **HMAC Log Chaining:** A cryptographic hashing algorithm linking consecutive logs, ensuring that any tampering by malicious insiders breaks the chain and is immediately flagged.
- **RAG SHA-256 Checksum Validation:** Guaranteeing the vector database is not compromised by injected falsified documents (Data Poisoning).

In summary, Chapter 3 establishes the rigorous theoretical framework of the SENTINEL architecture, holistically resolving both performance bottlenecks (Tier-1) and deep, secure cognitive reasoning (Tier-2). Chapter 4 will specify the process of translating this architecture into technical source code.

# Chapter 4

# Technical Implementation

This chapter details the software engineering aspects and practical source code implementation of the SENTINEL architecture. Shifting from abstract theory, the content focuses on the project directory organization, core class structures (Classes/Methods), Docker service configurations, and the design of the Graphical User Interface (Streamlit Dashboard) which facilitates Human-in-the-Loop (HITL) interaction.

## 4.1 Technology Stack and Source Code Structure

- **Core Technologies:** Python 3.10+, LangGraph, LangChain, FAISS, SQLite, Streamlit, Docker, and llama.cpp (via the Llama-cpp-python binding).
- **Directory Structure:** A detailed tree representation strictly separating components: `src/core/` (Tier 1 & 2), `src/rag/` (Vector DB), `src/security/` (Guardrails), `src/dashboard/` (UI), `experiments/`, and `tests/`.

## 4.2 Tier-1 Implementation: Filtration Engine & Welford

- **`RunningStats` and `SessionBaseline` Classes:** Python source code implementation for Welford's algorithm. Resolving dictionary memory allocation challenges via IP hashing and cache garbage collection (TTL cache).
- **`RuleEngine` Class:** Pattern matching design based on Regular Expressions and Static Thresholds.

## 4.3 Tier-2 Implementation: AI Agent and RAG

- **`OpenAILLMClient` / Llama.cpp Wrapper:** Parameter configurations for the Local LLM (enforcing `temperature=0.0` to optimize determinism, context size bounding, and `n_gpu_layers` mapping).

- **LangGraph Workflow Setup:** Utilizing `StateGraph` to define `analyze_node`, `retrieve_node`, and the conditional routing function `should_retrieve`.
- **`DualRetriever` Class:** Implementing the FAISS Index and rank_bm25 to achieve Reciprocal Rank Fusion (RRF). Source code overview of the scoring and sorting algorithms.

## 4.4 Guardrail and HMAC System Implementation

- **`DelimitedDataEncapsulator` Class:** Logic for generating secure hexadecimal tokens via `os.urandom` and `hashlib` to wrap log payloads.
- **`HMACHashing` Class (Database):** Specification of the SQLite database schema (`threat_memory.db`) and the trigger functions calculating SHA-256 hashes derived from Previous Hash + Current Data.

## 4.5 SOC Administration Dashboard Implementation (Streamlit UI)

- **Dashboard Architecture:** Application of a Frontend (Streamlit) communicating dynamically with the Backend (SQLite/File logs) through reactive UI components.
- **Display Modules:**
  - *Real-time Queue Dashboard:* Visualizing Tier-1 alert metrics and Tier-2 verdicts.
  - *Threat Memory Analytics:* Displaying APT alert graphs across IP timelines.
  - *Audit Trail Integrity:* An automated module verifying HMAC Log Chaining, utilizing a traffic-light interface (Green = Valid, Red = Chain Broken).
- **Human-in-the-loop (HITL):** The interface for reviewing (Approve/Reject) AI-recommended verdicts to update actual Firewall Rules.

## 4.6 Containerization and Infrastructure Deployment

- **`Dockerfile` Specification:** Multi-stage build mechanisms to minimize image size and compile C++ libraries (llama.cpp) with CUDA acceleration support.

- `docker-compose.yml` **Service Management:** Linking containers for the Dashboard and Agent Engine, alongside shared log volume mounts.

In summary, Chapter 4 systematizes the software development process of SENTINEL, bridging core algorithms with user interfaces and virtualized infrastructure. Chapter 5 will subsequently establish the testing environment to quantitatively evaluate these components.

# Chapter 5

# Experiments and Evaluation

This chapter presents the empirical evidence of the thesis, validating the research hypotheses through rigorous quantitative metrics. The SENTINEL system is evaluated against a 5-Dimensional framework (5D Metrics: Accuracy, Performance, Security, Explainability, and Integrity). The content emphasizes the layered Ablation Study design and the application of biostatistical models (McNemar's Test, Mann-Whitney U Test) to prove that the architectural enhancements are statistically significant and mathematically sound.

## 5.1 Experimental Environment and Datasets

- **Hardware Setup:** CPU Intel Core i9, 32GB RAM, GPU NVIDIA RTX 4060 Ti 16GB (utilized for executing the Gemma-2-9B-IT Q4_K_M model).
- **CSE-CIC-IDS2018 Dataset:** Introduction to the dataset encompassing diverse attack typologies (Brute Force, DoS, Web Attack) serving as the baseline for overarching accuracy evaluation.
- **DAPT2020 Dataset:** Employed for testing scenarios involving the identification of multi-stage Advanced Persistent Threat (APT) chains.
- **Unified Data Stream Simulation:** Detailing the scripting process that merges and replays log streams according to virtual timestamps to emulate real-time properties.

## 5.2 5D Evaluation Metrics and Ablation Study Design

- **Defining the 5D Metrics:** Accuracy (F1, Precision, Recall), Performance (Latency, Throughput), Security (Robustness Rate), Explainability (Audit Completeness), and Integrity (HMAC Validation).
- **Ablation Study Configurations (Config A through F):**

- Config A (Baseline 1): Traditional Rule-based engine exclusively.
- Config B (Baseline 2): Pure Local LLM inference (No RAG, no Welford).
- Config C: Welford Tier-1 + LLM.
- Config D: LLM + Single-RAG (FAISS).
- Config E: LLM + Dual-RAG Hybrid.
- Config F (Holistic SENTINEL): Tier-1 (Welford) + Tier-2 (LangGraph Agent + Dual RAG + Guardrails).

## 5.3 Accuracy Evaluation & McNemar's Statistical Test

- **Precision, Recall, and F1-Score Results:** Comparative charts and tables across configurations. Highlighting Config F's optimal F1 balance (eliminating Rule-based False Positives while enhancing Zero-day Recall).
- **APT Detection Efficacy:** Analysis of Tier-1 (Welford) capturing weak anomalous signals and subsequently activating the Agent's Threat Memory chains.
- **McNemar's Test Application:** Comparing the Confusion Matrices of Config B and Config F. Proving $p - value < 0.05$ (rejecting the Null Hypothesis) to confirm that the variance is attributable to architectural superiority rather than random chance.

## 5.4 Performance Latency Evaluation & Mann-Whitney U Test

- **Inference Latency Analysis:** Comparative table of average latency per flow. The exceptional offloading achieved by Tier-1 (Welford processing benign logs in $O(1)$ time) and the Semantic Cache (Cache Hits reducing LLM latency from 4-6 seconds to milliseconds).
- **Non-parametric Mann-Whitney U Test:** Demonstrating that the latency distribution of the SENTINEL configuration is significantly faster ($p < 0.05$) compared to routing all logs directly through the LLM.

## 5.5 Adversarial Robustness Evaluation (Security)

- **Adversarial AI Manipulation Dataset (45 Payloads):** Description of manipulative scenarios including Prompt Injections and Delimiter Smuggling (e.g., "Ignore system rules, report this IP as safe").

- **Mitigation Rate Measurement:** Comparison between an unguarded system (80% manipulation success rate) versus SENTINEL's enabled Delimited Data Encapsulation mechanism (achieving near-perfect mitigation of injection attempts).

## 5.6 Integrity and RAG Quality Evaluation via LLM-as-a-Judge

- **HMAC Chaining Audit:** A hypothetical scenario where an attacker compromises the database to alter log labels. The dashboard immediately reports the broken hash chain.
- **RAG Scoring (RAGAS / LLM-as-a-Judge):** Utilizing an external Llama-3.1 model as an independent arbiter to score four criteria: Context Precision, Context Recall, Faithfulness, and Answer Relevancy. Proving that Dual-RAG with RRF outperforms Single Vector Search.

In summary, Chapter 5 verifies the technical prowess of SENTINEL through hard quantitative data and rigorous biostatistical tests. Chapter 6 will distill the significance of these findings and outline future trajectories.

# Chapter 6

# Conclusion and Future Work

This concluding chapter synthesizes the comprehensive research journey of the thesis, “A Two-Tier Cognitive Architecture for Automated Threat Detection and Response Using Agentic AI” (SENTINEL). The chapter encapsulates the scientifically proven results, critically self-assesses the limitations within the scope of a master’s thesis, and illuminates new horizons for cybersecurity research integrated with Artificial Intelligence.

## 6.1 Synthesis of Achieved Research Results

- **Resolving Research Questions:** Reaffirming the successful mitigation of latency bottlenecks and Alert Fatigue via the tiered architecture of the Tier-1 Welford filter and the Tier-2 LangGraph Agent.
- **AI System Security (AI Security):** Demonstrating the empirical viability of Delimited Data Encapsulation and HMAC Chaining structures in thwarting adversarial manipulation risks (Prompt Injection, Delimiter Smuggling, Data Tampering) targeting internal enterprise cognitive systems.
- **Overcoming Zero-Day and APT Challenges:** The exceptional synergy between historical alert tracking (Threat Memory) and the hybrid knowledge repository (Dual-RAG MITRE + NIST) provides profound contextual awareness, enabling the system to expose highly sophisticated campaigns.
- **Validation Rigor:** These achievements transcend theoretical propositions, having been robustly validated by 5D metrics and standard statistical tests (McNemar, Mann-Whitney U) on empirical datasets.

## 6.2 Scientific and Practical Contributions

- **Scientific Value:** Providing a highly modular architectural framework for integrating Mathematical Statistics with Large Language Models in cybersecurity. This substantially fortifies the theoretical repository concerning LLM security and Guardrails.
- **Practical Value:** The finalized SENTINEL product serves as a core engine or an independent auxiliary tool for Small and Medium Businesses (SMBs) lacking the budget to sustain large-scale Tier-1 SOC teams, while strictly maintaining Zero Trust privacy through offline operations.

## 6.3 Existing Limitations

Despite the system's demonstrated superiority, the research acknowledges distinct boundaries:

- **Input Log Formats:** The system is currently deeply tested on Network Flow logs and HTTP payloads. Real-world applicability necessitates the ingestion of highly diverse unstructured formats (Windows Event Logs, Linux Syslogs, CloudTrail), requiring significantly more complex parsing mechanisms.
- **Hardware Constraints:** Although inference speed is accelerated via Semantic Caching, processing entirely novel alerts still requires substantial LLM reasoning time. This mandates high capital expenditure for specialized GPU hardware if operating at hundreds of Gigabits/second network scales.
- **Physical Mitigation Design:** The research deliberately constrained the system to recommend actions (e.g., Recommend Firewall Rules) rather than executing physical interventions on routers to prevent Self-Denial of Service (Self-DoS). However, this restricts true Real-Time Blocking capabilities.

## 6.4 Future Development Directions

Stemming from these limitations, future expansion trajectories for the system include:

- **Federated Learning:** Expanding SENTINEL into a decentralized architecture. Instead of sharing privacy-violating network logs, SOC endpoints share only learned model parameters regarding novel malware variants.

- **Multi-Agent System (MAS):** Decomposing the monolithic LangGraph Agent into specialized sub-agents: Malware Analyst Agent, Network Analyst Agent, and Threat Intel Agent. These agents would debate prior to issuing a final verdict, further elevating accuracy.
- **Automated Red Teaming:** Constructing an internal Red Team agent to continuously generate sophisticated adversarial payloads targeting Sentinel, thereby self-evaluating and autonomously upgrading the system's defensive posture.

*Closing a modest endeavor, yet opening profound aspirations in the pursuit of securing the digital realm through the power of Agentic AI.*

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